**Software Requirements Specification (SRS): Project Dalupotha**

**Version:** 1.0

## 1. Introduction

**Purpose of the system** The Dalupotha System is a centralized, digital ERP ecosystem designed to automate and manage the end-to-end supply chain of green tea leaves. It streamlines the collection process from Small Holders, optimizes field logistics for Transport Agents, and provides comprehensive factory management tools for financial tracking, inventory, and supplier services.

**Scope** The system encompasses an offline-first mobile application for field collection with IoT weight and GPS integration, a web-based management portal for factory operations (stores, brought leaf registry, deductions), and a self-service portal for Small Holders. It manages the workflows for advances, balance payments, fertilizer distribution, and transport requests. It does not cover the internal mechanical automation of tea processing machinery.

**Definitions and abbreviations**

- **TA:** Transport Agent
- **EXT:** Extension Officer
- **MG:** Manager / Extension Manager
- **SK:** Store Keeper
- **ST/FT:** Office Staff / Factory Staff
- **TRI:** Tea Research Institute
- **IoT:** Internet of Things
- **OTP:** One-Time Password

**Intended audience** This document is intended for project managers, lead software engineers, backend and frontend developers, QA engineers, system architects, and factory management stakeholders.

**System overview**

Dalupotha acts as the central nervous system for the tea factory. It connects mobile data captured in remote estates via Transport Agents to a central cloud database. This data instantly feeds into dashboards used by office staff to calculate daily brought leaf totals, deductions, and calculate payouts, while Small Holders access a simplified portal to view their financial standing and request services.

## 2. Overall Description

**Business context** Tea factories rely on thousands of independent Small Holders. Managing the physical collection, quality assessment, financial advances, and supply distribution manually leads to high operational friction and revenue leakage. Digitizing this workflow is critical to maintaining supplier loyalty and operational profitability.

**Product perspective** Dalupotha is designed as a modular, plug-and-play solution. Rather than a standalone silo, it is architected to seamlessly plug into broader organizational infrastructures, natively interfacing with dedicated accounting engines like Ginum for financial ledgers and PirisaHR for managing factory and extension staff payroll.

**User classes**

- **Small Holders:** Service receivers tracking deliveries and finances.
- **Transport Agents (TA):** Field operators capturing weights and logistics.
- **Factory/Office Staff (FT/ST):** Administrators processing daily registry and quality data.
- **Store Keepers (SK):** Inventory managers for fertilizer and bags.
- **Extension Officers/Managers (EXT/MG):** Approvers, monitors, and advisory providers.

**Operating environment**

- **Mobile Clients:** Android/iOS devices for TAs and Small Holders.
- **Web Clients:** Modern web browsers (Chrome, Edge, Safari) for factory staff.
- **Backend & Infrastructure:** Microservices deployed across scalable cloud environments such as Digital Ocean, Oracle Cloud Infrastructure (OCI), or Microsoft Azure, utilizing Supabase for real-time data sync and offline mobile capabilities.

**Constraints**

- Strict requirement for robust offline data caching due to poor cellular connectivity in remote tea estates.
- Hardware must endure harsh weather and possess high battery life for full-day field operations.

**Assumptions & dependencies**

- Bluetooth APIs for the hanging scales are open and accessible.
- Small Holders have access to basic mobile connectivity to receive OTP SMS messages.

## 3. Functional Requirements

### 3.1 Small Holder Registration & Profile Management

- **Input:** Mobile number, OTP, Name, Passbook No., Address, Land Name, live GPS coordinates, In charge No.
- **Processing:** The system validates the OTP via an SMS gateway. Upon success, it geotags the user's location, verifies the In charge No., and creates a secure profile in the database.
- **Output:** Successful registration confirmation; Small Holder profile generated with access credentials.
- **Error handling:** If OTP is invalid or expired, prompt retry. If GPS cannot be acquired, notify the user to move to an open area before finalizing registration.

### 3.2 Automated Green Leaf Collection (Transport Agent App)

- **Input:** Supplier ID (via QR/RFID or manual entry), Gross Weight (pulled via Bluetooth from IoT scale), current GPS coordinates.
- **Processing:** The mobile app pairs with the scale, captures the exact weight, associates it with the Supplier ID and GPS location. The data is cached locally (if offline) and queued for synchronization to the Supabase backend once the network is restored.
- **Output:** Digital collection receipt generated; Brought Leaf Register updated on the central server.
- **Error handling:** If the Bluetooth connection drops, prompt the TA to reconnect or allow a highly audited manual override. If sync fails, retain data in persistent local storage and retry exponentially.

### 3.3 Financial Advance Management

- **Input:** Advance amount requested by Small Holder or initiated by EXT/TA.
- **Processing:** The system evaluates the request against the supplier's current green leaf supply history and existing debts. It routes the request to the Manager/Extension Officer for approval. Once approved, the data triggers a ledger update via integration with the Ginum accounting module.
- **Output:** Status update on the Small Holder dashboard; notification of fund disbursement.
- **Error handling:** If the requested amount exceeds the allowable threshold based on supply history, automatically flag the request and display a "Limit Exceeded" warning.

### 3.4 Stores & Inventory Management (Fertilizer/Leaf Bags)

- **Input:** Quantity of fertilizer or leaf bags requested; Store Keeper dispatch confirmation.
- **Processing:** The system checks current inventory levels. Upon SK dispatch approval, the system deducts the stock from the warehouse inventory and applies the cost as a debt to the Small Holder's account for future deduction from balance payments.
- **Output:** Updated inventory ledger; updated Small Holder debt profile.
- **Error handling:** If stock is insufficient, block the approval and notify the Store Keeper to initiate a restock workflow.

## 4. Non-Functional Requirements

### Performance

- The system must support **500 concurrent users** during peak collection and payment processing hours.
- Order processing time (e.g., syncing a leaf collection record or processing an advance request) must be **< 3 seconds** under normal network conditions.

### Security

- Implementation of strict **Role-based access control (RBAC)** ensuring TAs, SKs, and Small Holders only access authorized data modules.
- All data transmission between mobile, web clients, and cloud servers must be fully encrypted using **HTTPS (TLS 1.2+)**.
- Comprehensive **Audit logs for transactions**, specifically tracking manual weight overrides, advance approvals, and balance payment deductions.

### Reliability

- The system architecture must guarantee **99.5% uptime**, leveraging load balancing across the cloud infrastructure.
- **Automatic daily backups** of all database records, user profiles, and transaction histories to a secure, redundant storage bucket.

### Usability

- **Simple UI for low-technical staff:** The mobile app for TAs and the portal for Small Holders must rely heavily on visual cues, large touch targets, and minimal text entry to accommodate varying digital literacy.

- **Training time < 2 days:** The system workflows must be intuitive enough that a newly hired Transport Agent or Store Keeper is fully operational within two days of training.

**Scalability**

- **Must handle expansion to multiple warehouses:** The database schema and tenant architecture must support scaling to accommodate additional factory branches, multi-region stores, and expanded vehicle fleets without requiring a core code rewrite.

**5. External Interface Requirements**

- **Hardware Interfaces:**
  - *IoT Hanging Scales:* Mobile application requires Bluetooth Low Energy (BLE) protocols to interface directly with ruggedized digital scales to parse continuous hexadecimal weight data streams.
  - *Mobile GPS Sensors:* Direct access to the device's native location services hardware to capture precise lat/long coordinates.
- **Software Interfaces:**
  - *Accounting System API:* Secure REST API connection to the **Ginum** platform to push daily balance payments, advances, and retrieve ledger statuses.
  - *HR System API:* Interface with the **PirisaHR** module to validate internal staff (TA, EXT, FT) credentials, shift tracking, and performance metrics.
- **Communication Interfaces:**
  - *SMS Gateway API:* Integration with a local telecommunications provider to dispatch OTPs for Small Holder registration and critical alerts.
  - *WebSockets/Real-time DB:* Utilization of protocols (via Supabase) to ensure the management web dashboard updates in real-time as TAs sync data from the field.